

Selecting Informative Mismatches for Same-State Counterfactual Evaluation

Can a Low-Cost Mock Rank Predict Physics-Level Counterfactual Value?

Seonhye Gu
AI-based Surgical Robot Innovation Lab
Yonsei University College of Medicine

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Exception-aware Physical AI Research Program · Paper 002

Pre-results manuscript · Frozen before confirmatory GPU execution

Status and Disclaimer

This document is a **pre-results discussion draft**. The research question, hypotheses, experimental contract, operational gate, exclusion rules, and statistical analysis were frozen before confirmatory Isaac execution under tag **paper002-prereg-v0.3**. Placeholder result tables intentionally contain no confirmatory outcomes.

The study uses a scripted reach task and a gain-scale visibility proxy in simulation. It does **not** evaluate patients, clinical workflows, autonomous surgery, learned recovery policies, real surgical-image occlusion, real hardware transfer, or medical safety. The manuscript is not peer reviewed and should not be cited as an experimental result until confirmatory execution and reporting are complete.

Author and affiliation: Seonhye Gu · AI-based Surgical Robot Innovation Lab · Yonsei University College of Medicine

Public artifact boundary: <https://github.com/higuseonhye/research-os>

Freeze tag: paper002-prereg-v0.3 · **Config:** sandbox_v0.4.yaml

Abstract

Physical-AI systems are commonly evaluated under researcher-selected disturbances, yet many candidate failures are redundant, trivial, or uninformative about the value of alternative responses. Automatic curriculum and environment-generation methods can expand the space of candidate disturbances, but evaluating every candidate in high-fidelity physics simulation is computationally expensive. We study whether a low-cost mock evaluator can identify candidate mismatches that remain informative under high-fidelity Isaac simulation.

Under a frozen occlusion contract, deterministic rule-based and large-language-model-based planners generate candidate perturbations. Each candidate receives a continuous mock counterfactual score across five seeds, and candidates are ranked within planner before physics evaluation. A frozen transfer function maps mock parameters to Isaac parameters. Starting from a fixed robot and world state, paired CONTINUE and REPLAN responses are then executed over eight Isaac seeds. The primary confirmatory analysis tests rank transfer and top-versus-bottom enrichment on the rule-planner export set. A secondary hierarchical analysis evaluates whether the LLM planner satisfies both an absolute correlation floor and a non-inferiority criterion relative to the rule planner.

This study is positioned as a validation and experiment-selection study rather than a policy-learning or curriculum-training study. It asks which mismatches are worth evaluating, not whether generated curricula improve a learned policy. The manuscript specifies the research question, related work, frozen methods, statistical analysis, operational gate, and preregistered result structure before confirmatory GPU execution.

Keywords: Physical AI, informative mismatch, counterfactual evaluation, experiment selection, proxy validation, recoverability, robotic simulation.

1 Introduction

Physical-AI systems are usually developed and evaluated under a limited set of disturbances selected by researchers. Such disturbances can demonstrate that a system fails, but they do not necessarily reveal whether the failure is useful for comparing alternative responses, diagnosing limitations, or constructing subsequent experiments.

We define an *informative mismatch* as a perturbation that produces a reliable, task-relevant difference between two prespecified responses from the same initial state. Informativeness is therefore distinct from raw difficulty. A perturbation can make a task harder without helping distinguish whether replanning is preferable to continuing the existing plan.

PAPER 001 established the same-state counterfactual fork used here. At a frozen world and robot state S , a mismatch is introduced and two response modes are compared:

$$S \xrightarrow{\text{mismatch}} \begin{cases} \text{CONTINUE,} \\ \text{REPLAN.} \end{cases} \quad (1)$$

PAPER 002 addresses the preceding selection problem: given many possible mismatches, can a low-cost approximation identify which candidates are likely to be informative when evaluated in physics?

Three questions are often conflated:

1. Can a system generate valid perturbation specifications?
2. Can a low-cost evaluator rank those specifications?
3. Do the selected perturbations remain informative under physics simulation?

The primary contribution concerns the second and third questions. Candidate generation is treated as an experimental factor, not as sufficient evidence of curriculum quality.

1.1 Problem definition

At a fixed post-mismatch execution state S , Paper 001 provides the measurement ruler: response-conditioned terminal outcomes under matched replay. Paper 002 studies the upstream selection problem. Given a candidate set $\mathcal{C}$ of mismatch specifications, the goal is to determine whether a low-cost score $\bar{M}_i$ can prioritize specifications with high physics-level counterfactual value I_i .

This formulation deliberately separates *generation*, *ranking*, and *physics validation*. A generator may produce diverse or plausible specifications without providing a valid ranking. Conversely, a simple rule generator can still be useful if a cheap evaluator reliably selects experiments that expose a meaningful difference between CONTINUE and REPLAN.

1.2 Research-program position

The study forms the front end of a broader exception-aware physical-AI program:

1. **Paper 002:** Which mismatch should be evaluated?
2. **Paper 001:** How should recoverability be measured from the same state?
3. **Paper 003:** How can recovery be improved or latent mismatch be detected?
4. **Paper 004:** When should the system revise its world model?

Accordingly, PAPER 002 is best understood as an experiment-selection or evaluation-curriculum study. It does not yet optimize a recovery policy or update a world model.

1.3 Research question

Under a frozen occlusion contract, does continuous mock-based ranking predict same-state counterfactual informativeness in Isaac simulation on the frozen export set, and does an LLM goal planner meet prespecified adequacy criteria relative to a deterministic rule planner?

The ordered claims are:

1. **Proxy validity:** higher mock scores correspond to higher Isaac counterfactual scores.
2. **Selection utility:** top-ranked candidates are enriched for informative Isaac cases relative to bottom-ranked candidates.
3. **Planner adequacy:** the LLM planner satisfies an absolute floor and is non-inferior to the rule planner within the prespecified margin.

1.4 Contributions

This study makes four bounded contributions:

1. A proxy-validation formulation for selecting physical-AI failures.
2. A same-state informativeness target defined by paired CONTINUE/REPLAN comparisons.
3. A frozen mock-to-physics transfer and export contract.
4. A controlled LLM-versus-rule comparison using an absolute floor and a non-inferiority margin.

2 Related Work

2.1 Automatic curriculum learning and prioritized sampling

Curriculum learning organizes examples or tasks so that a learner encounters a useful progression rather than a uniformly sampled set. In reinforcement learning, the value of a level depends on the current policy. Prioritized Level Replay (PLR) formalizes this by prioritizing levels with greater estimated learning potential, improving sample efficiency and generalization in procedurally generated environments [1]. Earlier prioritized replay work similarly showed that transitions need not be sampled uniformly when their learning value differs [2].

PAPER 002 adopts the general distinction between available candidates and useful candidates, but its outcome is not policy improvement. It selects *experiments* for high-fidelity evaluation. The target is a physics-derived counterfactual value rather than temporal-difference error, regret, or expected gradient utility.

2.2 Unsupervised environment design and open-endedness

POET and Enhanced POET jointly generate environmental challenges and optimize agents, allowing solutions to transfer across an expanding population of tasks [3, 4]. Unsupervised environment-design methods such as PAIRED use relative regret to create environments that expose differences between agents and induce adaptive curricula [5]. More recent work expands this view to joint task-level curricula [8].

These methods ask whether generated environments improve learning or produce increasingly capable agents. In contrast, PAPER 002 freezes the response policies and asks whether a cheap proxy preserves the ordering of candidate perturbations under a more expensive physics evaluator.

2.3 LLM-generated robotic tasks

GenSim uses large language models to generate executable robotic simulation tasks and demonstrations, with downstream evaluation based on multitask policy training and task-level generalization [6]. RoboGen similarly uses generative models to propose tasks, construct scenes, and generate training supervision for large-scale robot learning [7].

PAPER 002 assigns the LLM a narrower role. It proposes parameters within a frozen perturbation schema. The LLM cannot change the evaluator, simulator mapping, response policies, hypothesis thresholds, or analysis plan. The study therefore evaluates an LLM as a constrained candidate proposer rather than as an autonomous experiment designer.

2.4 Simulation fidelity and proxy validation

Most simulation-transfer research studies whether policies, dynamics, or observations transfer across domains. Here, the object of transfer is the *ordering of candidate experiments*. The mock evaluator is not required to reproduce exact Isaac trajectories. It is useful if it preserves enough relative information to prioritize candidates for expensive evaluation.

This motivates rank association and top-versus-bottom enrichment as primary outcomes rather than absolute trajectory agreement.

2.5 Counterfactual evaluation, recoverability, and evaluation curricula

Robotic robustness is often evaluated through task success under perturbations, replanning performance, or intervention recovery. Comparisons can be confounded when trajectories reach different states before the intervention. The same-state fork reduces this source of variation by holding the pre-mismatch state fixed.

The present study does not claim to solve recovery. It validates a method for selecting mismatches that reveal differences between recovery responses. In this sense, it constructs an *evaluation curriculum*: an ordered set of experiments selected for measurement value rather than a training curriculum selected for policy optimization.

3 Methods

3.1 Study design

The study uses a staged mixed design with the following factors:

- **Planner:** rule or LLM.
- **Dreamer:** Gaussian or diffusion candidate-generation process.
- **Export tier:** top or bottom according to the consensus mock score.

Candidate generation occurs before Isaac evaluation. Export is performed within planner: the top 10 and bottom 10 candidates are selected after pooling dreamers, yielding 20 Isaac specifications per planner and 40 total.

The confirmatory populations follow the frozen preregistration:

- H1 and H2: rule-planner export set, nominal $n = 20$.
- H3: rule-planner $n = 20$ versus LLM-planner $n = 20$.
- Pooled analyses: descriptive only unless explicitly labeled otherwise.

3.2 Candidate specification and occlusion contract

Each candidate is represented by a canonical specification containing:

- target shift in meters;
- occlusion onset step;
- occlusion gain;
- planner, dreamer, and seed metadata;
- a canonical specification identifier.

All candidates must satisfy a frozen occlusion contract:

1. only the prespecified visual-observation component is modified;
2. robot dynamics, controller parameters, reward, and base task remain unchanged;
3. perturbation onset follows the frozen mismatch timing;
4. values remain within simulator-safe bounds;
5. invalid or unparsable outputs are rejected before scoring;
6. rejection and duplication counts are retained by planner.

3.3 Rule and LLM planners

The rule planner generates candidates from deterministic tier-constrained parameter ranges. Random seeds, enumeration order, validation, and replacement behavior are frozen.

The LLM planner receives the canonical JSON schema, allowed parameter ranges, task description, and generation constraints. It does not receive Isaac outcomes, mock thresholds, exported membership, hypothesis thresholds, or rule-planner scores. The model identifier, prompt, decoding parameters, retry count, parser, and repair policy are frozen and logged.

3.4 Continuous mock score

For candidate i and mock seed s , the primary mock score is:

$$M_{is} = (\mathbb{I}[\text{REPLAN ok}] - \mathbb{I}[\text{CONTINUE ok}]) + \text{clip}\left(\frac{d_{C,is} - d_{R,is}}{4\tau}, 0, 1\right) + 0.25\mathbb{I}[\text{violation lift}], \quad (2)$$

where $d_{C,is}$ and $d_{R,is}$ are branch-specific terminal or residual distances and $\tau = 0.02$ m is the frozen mock tolerance.

The binary mock label is secondary:

$$\hat{Y}_{is}^{\text{mock}} = \mathbb{I}[M_{is} \geq 0.5]. \quad (3)$$

The confirmatory consensus score is the median over seeds 42–46:

$$\bar{M}_i = \text{median}_{s \in \{42,43,44,45,46\}} M_{is}. \quad (4)$$

Spearman analyses use the continuous score $\bar{M}_i$, not the binary label.

3.5 Deduplication and consensus merge

Candidates are matched by the canonical key:

$$(\text{round}(\text{shift_m}, 4), \text{onset_step}, \text{round}(\text{occlusion_gain}, 3)). \quad (5)$$

Exact duplicates are resolved using the frozen retention policy. Consensus records are constructed only from seeds 42–46. Seed 43 may be used separately for engineering smoke testing or sensitivity analysis, but smoke artifacts cannot enter the confirmatory ranking.

3.6 Within-planner export

For each planner:

1. pool candidates across dreamers;
2. sort by descending $\bar{M}_i$;
3. break ties lexicographically by specification identifier;
4. export the top 10 and bottom 10 candidates;
5. write the frozen manifest;
6. record its SHA256 checksum before Isaac execution.

No candidate specification, mapping, threshold, or ranking rule may be changed after the manifest checksum is recorded.

3.7 Mock-to-Isaac transfer

The frozen transfer is shown in Table 1.

The Isaac runner is the frozen `study1d` pipeline using the `gain_scale_flag_v0.1` proxy at level 1.

Table 1: Frozen mock-to-Isaac transfer contract.

Mock field	Isaac field	Transfer rule
<code>occlusion_gain</code>	<code>visibility_fraction</code>	$\text{clip}(1 - g, 0.05, 1)$
<code>onset_step</code>	perturbation start	identical integer step
<code>shift_m</code>	target shift in $Y+$	direct transfer
<code>tier</code>	metadata	no simulator effect

3.8 Isaac same-state counterfactual evaluation

All evaluations begin from the fixed state S at control step 20, following the PAPER 001 state contract. Each candidate is evaluated under:

- CONTINUE: continue the pre-mismatch plan;
- REPLAN: replan immediately at mismatch onset (`REPLAN_d0`).

Each specification is evaluated over eight Isaac seeds, indexed 0–7. A specification is valid when at least four paired seeds complete successfully.

Let D_{is} be the seed-level continuous counterfactual margin, computed using the frozen branch-difference definition. The candidate-level continuous Isaac score is:

$$I_i = \text{median}_{s \in V_i} D_{is}, \quad (6)$$

where V_i is the set of valid paired seeds.

For eight valid seeds, the binary candidate label is:

$$Y_i = \mathbb{1} \left[\sum_{s=1}^8 Y_{is} \geq 5 \right]. \quad (7)$$

When four to seven seeds are valid, the frozen majority-of-valid rule is applied. Specifications with fewer than four valid paired seeds are excluded from confirmatory analysis.

3.9 Staged execution and operational gate

Execution proceeds as follows:

1. freeze documentation, code, configuration, commit, and tag;
2. optional seed-43 engineering smoke;
3. generate LLM curricula over five seeds;
4. run rule and LLM mock evaluations over seeds 42–46;
5. merge consensus, export within planner, and freeze manifest checksums;
6. run the rule-planner Isaac leg;
7. apply the engineering operational gate;
8. run the LLM-planner Isaac leg if the operational gate passes;

9. evaluate H1, H2, and H3 after both Isaac legs.

The operational gate is not a hypothesis test. Continuation to the LLM Isaac leg requires:

- at least 90% of rule specifications start both branches;
- at least 80% of rule specifications have at least four valid seed pairs;
- no systematic transfer-contract or runner failure;
- non-degenerate CONTINUE/REPLAN informativeness across the 20 rule specifications.

If the gate fails, the study is reported as a feasibility study. The frozen export may not be tuned using rule outcomes.

3.10 Engineering smoke protocol

A seed-43 smoke run is permitted before confirmatory execution only for pipeline validation. Smoke results:

- do not enter consensus export;
- cannot change thresholds, weights, mappings, or exclusions;
- cannot be reused in the frozen export manifest;
- are stored in a separate manifest and labeled as engineering artifacts.

4 Hypotheses and Statistical Analysis

4.1 H1: rule-planner rank transfer

On the rule-planner export set:

$$\rho_S(\bar{M}_i, I_i) \geq 0.50. \quad (8)$$

H1 passes only if:

1. observed $\rho_S \geq 0.50$;
2. at least 18 rule candidates remain after exclusions;
3. the lower bound of the candidate-level 95% bootstrap confidence interval exceeds 0.25;
4. the preregistered permutation sanity check is reported.

The bootstrap uses 2,000 candidate-level resamples. The permutation analysis uses 1,000 shuffles and computes:

$$p_{\text{perm}} = \frac{1 + \sum_{b=1}^{1000} \mathbb{I}(|\rho_b| \geq |\rho_{\text{obs}}|)}{1001}. \quad (9)$$

The confirmatory claim is restricted to the frozen extreme export set. It is not global proxy validity over all generated candidates.

4.2 H2: rule-planner top-tier enrichment

For the rule planner:

$$\text{IR}_{\text{top}} = \frac{1}{10} \sum_{i \in T} Y_i, \quad \text{IR}_{\text{bottom}} = \frac{1}{10} \sum_{i \in B} Y_i. \quad (10)$$

H2 passes only if all conditions hold:

$$\text{IR}_{\text{top}} \geq 0.80, \quad (11)$$

$$\text{IR}_{\text{bottom}} \leq 0.50, \quad (12)$$

$$\text{IR}_{\text{top}} - \text{IR}_{\text{bottom}} \geq 0.40. \quad (13)$$

A one-sided exact binomial test against $p_0 = 0.50$ is reported as supporting inference, not as an additional pass requirement. For $n = 10$, 8/10 informative yields $p = 0.0547$, whereas 9/10 yields $p = 0.0107$.

4.3 H3: LLM adequacy and non-inferiority

H3 is evaluated after both planner legs and after the hierarchical H1 and H2 steps.

The absolute floor is:

$$\rho_{\text{LLM}} \geq 0.50. \quad (14)$$

The planner difference is:

$$\Delta_\rho = \rho_{\text{LLM}} - \rho_{\text{rule}}. \quad (15)$$

The LLM is non-inferior only when the lower bound of the 95% bootstrap confidence interval satisfies:

$$\text{LB}_{95\%}(\Delta_\rho) > -0.15. \quad (16)$$

Both H3a and H3b are required. Failure to establish non-inferiority is not interpreted as proof of inferiority.

4.4 Multiplicity

Hypotheses are interpreted hierarchically:

$$H1 \rightarrow H2 \rightarrow H3. \quad (17)$$

Pooled planner analyses, middle-tier analyses, alternative correlations, and disagreement analyses are descriptive or sensitivity analyses unless otherwise specified.

4.5 Missingness and exclusions

No missing Isaac outcome is imputed. Exclusions are categorized as:

- simulator initialization failure;
- state-restore failure;
- numerical instability;
- timeout;
- invalid candidate mapping;

- controller exception;
- corrupted output;
- other prespecified engineering failure.

Failure rates are reported by planner and export tier.

4.6 Sensitivity analyses

Prespecified Tier-B analyses include:

1. seed-43-only export;
2. binary mock label instead of continuous score;
3. Kendall’s τ instead of Spearman’s ρ ;
4. optional middle-ranked five candidates per planner;
5. pooled planner correlation reported descriptively.

5 Sample-Size Rationale

The confirmatory export contains 40 candidates: 20 rule and 20 LLM. H1 and H2 use the 20 rule candidates; H3 compares planner-specific correlations using 20 candidates per planner.

The sample size is constrained by high-fidelity simulation cost and targets a large rank association rather than small interaction effects. The design is not intended for fine-grained planner-by-dreamer interaction testing. The non-inferiority analysis is therefore a bounded confirmatory test. A simulation-based power analysis may be added before submission, but no post-result power calculation will replace effect estimates and confidence intervals.

Without the optional middle-ranked sample, conclusions must remain limited to the extreme export set.

6 Preregistered Result Structure

Table 2: T1. Candidate accounting.

Planner	Raw	Valid	Unique	Exported	Isaac-valid	Confirmatory
Rule	–	–	–	20	–	–
LLM	–	–	–	20	–	–
Total	–	–	–	40	–	–

7 Interpretation Rules

8 Limitations and Claim Boundaries

The study has six principal limitations:

Table 3: T2. Rank transfer.

Population	n	Spearman ρ	95% CI	Permutation p	Decision
Rule export set	—	—	—	—	—
LLM export set	—	—	—	—	—
Pooled, descriptive	—	—	—	—	—

Table 4: T3. Tier enrichment.

Planner	Group	n	Informative	IR	Exact 95% CI
Rule	Top	10	—	—	—
Rule	Bottom	10	—	—	—
LLM	Top	10	—	—	—
LLM	Bottom	10	—	—	—

1. It evaluates a single controlled perturbation family.
2. Extreme-group export may inflate apparent rank separation.
3. Planner-specific inference is based on small candidate sets.
4. Isaac simulation is not equivalent to real hardware or clinical operation.
5. LLM conclusions are tied to a frozen model, prompt, schema, and decoding configuration.
6. Informativeness is defined relative to a frozen CONTINUE/REPLAN pair.

The study may claim proxy rank validity and selection utility only within the tested contract and population. It may not claim improved policy learning, general physical-AI robustness, sim-to-real transfer, surgical safety, or general superiority of LLM planners.

9 Data, Code, and Reproducibility

The public repository records the frozen preregistration, method specification, analysis plan, configuration, JSON schema, export utilities, and analysis code. The confirmatory freeze point is:

`paper002-prereg-v0.3`

The final manuscript will report:

- commit and tag identifiers;
- manifest SHA256 checksums;
- simulator and dependency versions;
- raw-to-confirmatory candidate accounting;
- all exclusions and operational failures;
- deviations from the frozen protocol, if any.

Table 5: T4. Hypothesis decisions.

Hyp.	Criterion	Estimate	Decision
H1	$\rho_{\text{rule}} \geq 0.50$ and bootstrap lower bound > 0.25	–	–
H2	$\text{IR}_{\text{top}} \geq 0.80$, $\text{IR}_{\text{bottom}} \leq 0.50$, gap ≥ 0.40	–	–
H3a	$\rho_{\text{LLM}} \geq 0.50$	–	–
H3b	bootstrap lower bound for $\Delta_\rho > -0.15$	–	–

Table 6: Prespecified interpretation language.

Observed pattern	Permitted interpretation
H1 and H2 pass	Proxy-guided mismatch selection is supported on the frozen rule-planner export set.
H1, H2, and H3 pass	The constrained LLM proposer additionally meets the prespecified absolute and non-inferiority criteria.
H1 passes, H2 fails	Mock and Isaac scores are associated, but the top/bottom filter is not sufficiently useful under the frozen threshold.
H1 fails	Rank transfer is not validated under the tested contract; enrichment findings are descriptive.
H3a passes, H3b fails	The LLM reaches the absolute floor, but non-inferiority to the rule planner is not established.

10 Protocol Consistency with Paper 001

Paper 001 measured intervention-conditioned recoverability at a matched simulator execution state. Paper 002 reuses that state-restoration and response-fork logic as the high-fidelity judge. The connection is therefore operational rather than rhetorical: Paper 001 defines what counts as a response difference at S , while Paper 002 tests whether candidate mismatches worth measuring can be selected before full Isaac execution.

To keep the manuscripts consistent, both papers use bounded claims. “Same-state” denotes a replay-restored execution state satisfying the frozen quality-control contract, not byte-identical simulator memory. “Occlusion” in the current stage denotes a gain-scale visibility proxy, not geometric or image-space occlusion. “Informative” is relative to the frozen CONTINUE/REPLAN pair and does not imply clinical importance.

Pre-results Status

At the time of this manuscript freeze, confirmatory Paper 002 Isaac outcomes have not been used to define or revise the confirmatory protocol. Tables contain placeholders by design. The Introduction, Related Work, Methods, hypotheses, analysis plan, interpretation rules, and claim boundaries are intended to be fixed before GPU execution.

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